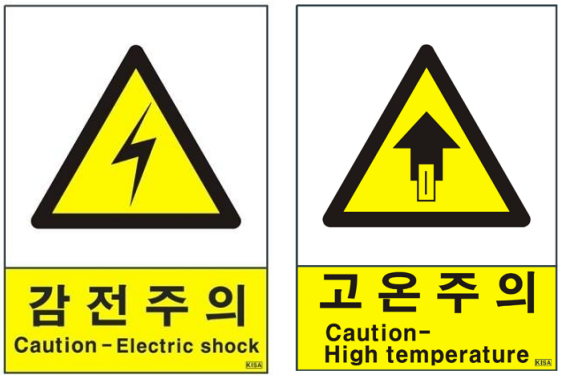
Photoelectric effect



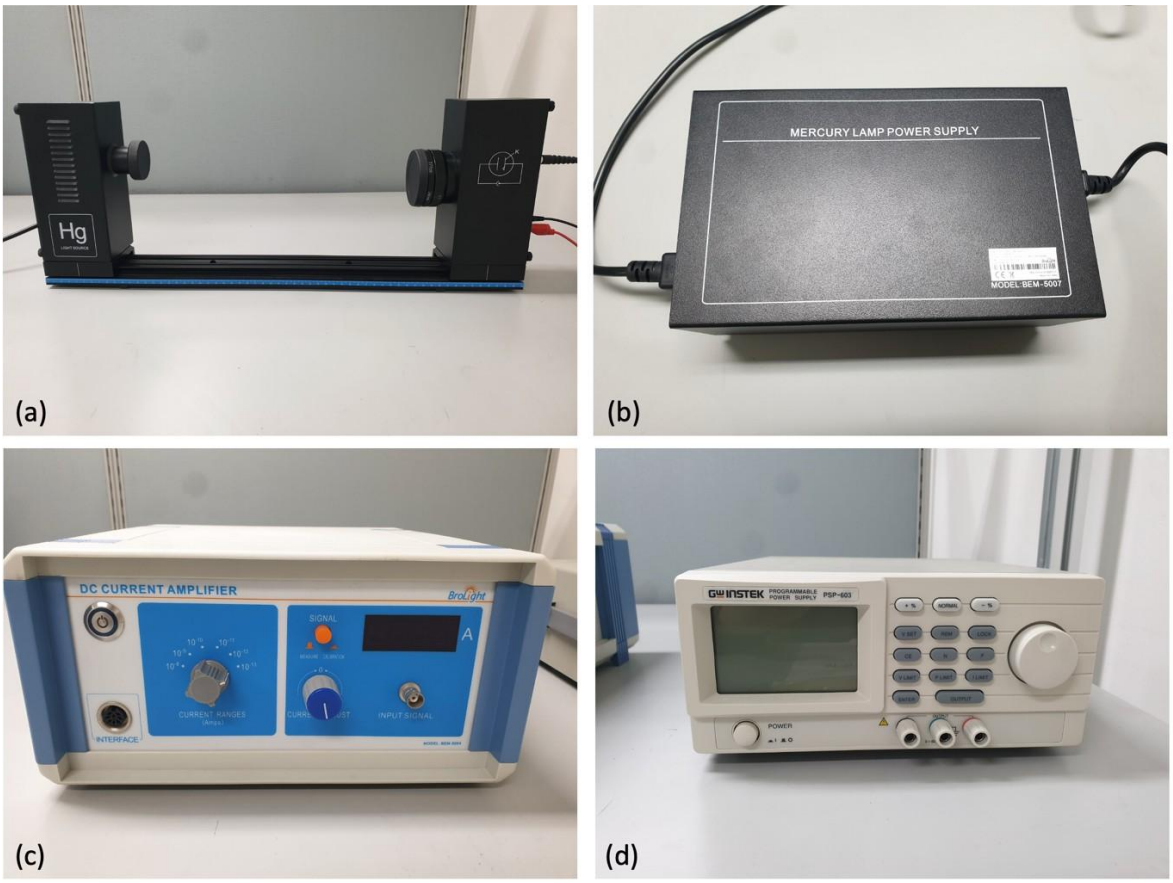


Fig. 1. (a) Lamp(left) and photodiode(right). (b) Lamp power supply.
(c) DC current amplifier. (d) Power supply.

EXPERIMENT OBJECTIVE

Observe particle-like nature of light.

INTRODUCTION

The photoelectric effect is the emission of electrons from the surface of a metal when electromagnetic radiation (such as visible or ultraviolet light) shines on the metal. At the time of its discovery, the classical wave model for light predicted that the energy of the emitted electrons would increase as the intensity (brightness) of the light increased. However, experiments showed that this was not the case. Instead, treating light as a particle (photon) provided a more consistent explanation of the observed behavior.

In this lab, you will study the effect of varying the light intensity and frequency on the energy of the emitted electrons and the magnitude of the photocurrent. You will also determine Planck's constant.

PREDICTIONS

1. If the light shining on the metal becomes brighter, will the kinetic energy of the emitted electrons increase?

2. If the light shining on the metal becomes brighter, will the number of emitted electrons per second (photocurrent) increase?

3. If the frequency of the light shining on the metal increases, will the number of emitted electrons per second (the photocurrent) increase?

4. If the frequency of the light shining on the metal increases, will the kinetic energy of emitted electrons increase?

5. Will all colors (frequencies) of light cause electrons to be emitted by the metal?

6. If light with the same frequency and intensity shines on two different metals with different work functions:
   a. Will both metals emit electrons?
   b. If both metals emit electrons, will these electrons have the same kinetic energy?
   c. Will the photocurrent be larger for the metal with the smaller work function?

SET UP

1. Mount the mercury lamp and the photodiode case on the track.

2. Connect the power cord from the Mercury Light Source enclosure into the receptacle labeled "POWER OUTPUT FOR MERCURY LAMP" on the Mercury Lamp Power Supply. Connect the Mercury Lamp Power Supply to an outlet.

3. Turn on the Mercury Lamp and let it warm up for at least 10 minutes. Leave the cover on the lamp to avoid looking directly into the lamp.

4. Do not connect any cords to the photodiode yet.

5. Connect the cables to the **photodiode**:

6. Connect the **BNC-to-BNC cable** from the port labeled **K** on the photodiode to the **BNC jack** on the DC Current Amplifier.

- Connect the **BNC-to-BNC cable** from the port labeled **K** on the photodiode to the **BNC jack** on the DC Current Amplifier.
- Connect the **red banana cable** from the port labeled **A** on the photodiode to the **black banana jack** on the Power Supply.
- Connect the **black banana cable** from the black banana jack on the photodiode to the **red banana jack** on the Power Supply.

※**Note:** For Experiments A and B, keep the connections as described above so that the applied voltage is opposite to the photocurrent direction (required for measuring the stopping voltage). For Experiment C, reverse the banana cable connections if needed to measure the current in the opposite polarity.

7. Turn on the DC current amplifier.

8. On the DC Current Amplifier, turn the **CURRENT RANGES** switch to **10^-13**. Press the **SIGNAL** button to enter calibration mode. Turn the **CURRENT ADJUST** knob until the meter reads 000 A.

9. Press the **SIGNAL** button again, and allow it to return to the out position for **MEASURE** mode.

10. On the Power Supply, press the **OUTPUT** button so that *Output* **ON** is displayed on the screen. Then press the **F** button to enable fine adjustment of the voltage in steps of **0.01 V**. If you want to adjust the voltage in **1.00 V**, Press the **N** button.

11. During the experiment, you will be changing the aperture and the filters.
    a. To change the aperture, pull out on the **Aperture Ring** and rotate it until it clicks into place.
    b. To change the filter, do not pull out; just rotate the **Filter Ring** until it clicks into the next position.

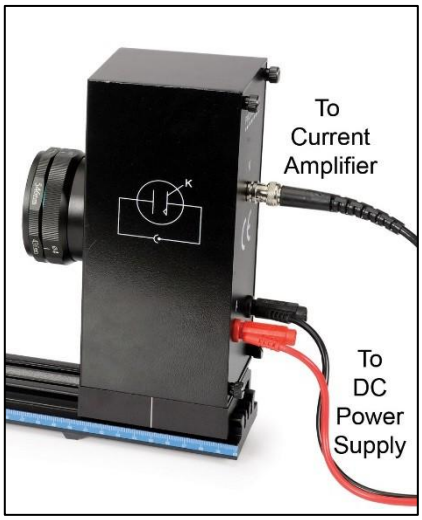


Fig. 2. Photodiode Connections

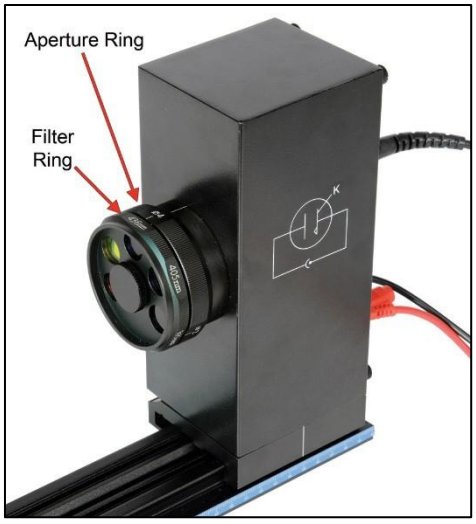


Fig.3. Filter Ring and Aperture Ring on Photodiode

A. STOPPING VOLTAGE vs. WAVELENGTH

1. Select the 2 mm aperture and the 365 nm filter on the photodiode.
2. Remove the Mercury Light Source cover.
3. Turn the right-hand knob to adjust the voltage until the current on the ammeter is zero.
4. Rotate the filter ring to the next longer wavelength filter and adjust the voltage until the current on the ammeter is zero.
5. Repeat the procedure for all the filters.
6. Put the cover back onto the Mercury Light Source.

B. STOPPING VOLTAGE vs. LIGHT INTENSITY

To investigate how light intensity affects the energy of the emitted electrons, the intensity is varied by changing the aperture while keeping the wavelength constant.

1. Select the 436 nm filter and 8 mm aperture on the photodiode.
2. Before remove the cover on the light source, adjust the **CURRENT ADJUST** knob until the meter reads 000 A.
3. Uncover the mercury light source and adjust the voltage until the current on the ammeter is zero. Record the stopping voltage.
4. With the cover on the mercury light source, change the aperture from the 8 mm to 4 mm aperture on the photodiode tube.
5. Repeat 1~3 for 4mm and 2mm aperture.

C. CURRENT vs LIGHT INTENSITY

1. With the cover on the mercury light source, select the aperture to 2 mm aperture on the photodiode tube and put on the 436 nm filter.
2. If you need to change the **CURRENT RANGES** switch on the DC current amplifier during the experiment, recalibrate it afterward.
3. Uncover the mercury light.
4. Adjust the voltage slowly from -3 V to +24 V in steps small enough to define a smooth curve. Each time you wait until the readings are stable.

※ Decide for yourself which range to measure and what step size to use in more detail.

5. Change the aperture to 4 mm. This makes the light brighter. Repeat the procedure.

ANALYSIS

1. Perform a linear fit to the data.
2. Calculate Planck's constant from the slope of the line.
3. Calculate the percent difference between the result and the accepted value.
4. From the y-intercept, find the work function for the metal.
5. Additionally, include any further graphs or analyses that you find relevant.

QUESTIONS

1. Which wavelength gave the emitted electrons the greatest kinetic energy?
    - What color is this?
2. From the x-intercept, find the cut off frequency.
    - What wavelength(color) does the cutoff frequency correspond to?

THEORY

The electrons in a metal are in potential energy wells (see Figure 1A). When light shines on the metal, the energy of the photon is absorbed by the electron and converted into potential energy and kinetic energy. If the photon has sufficient energy to raise the electron out of the potential well, the electron will be ejected from the surface of the metal with kinetic energy (K) equal to the difference between the photon energy (E) and the depth of the potential well (φ, the work function of the metal) (see Figure 1B).

$$K = E - \varphi \tag{1}$$

The energy of the photon is proportional to its frequency:

$$E = hf \tag{2}$$

where h denotes Planck's constant.

If the energy of a photon is equal to the work function, the electron makes it out of the well with no kinetic energy and thus will not actually leave the surface of the metal. If the energy of the photon is greater than the work function, the electron makes it out of the well with some kinetic energy and the electron leaves the surface of the metal. If the energy of the photon is less than the work function, the electron does not make it out of the well.

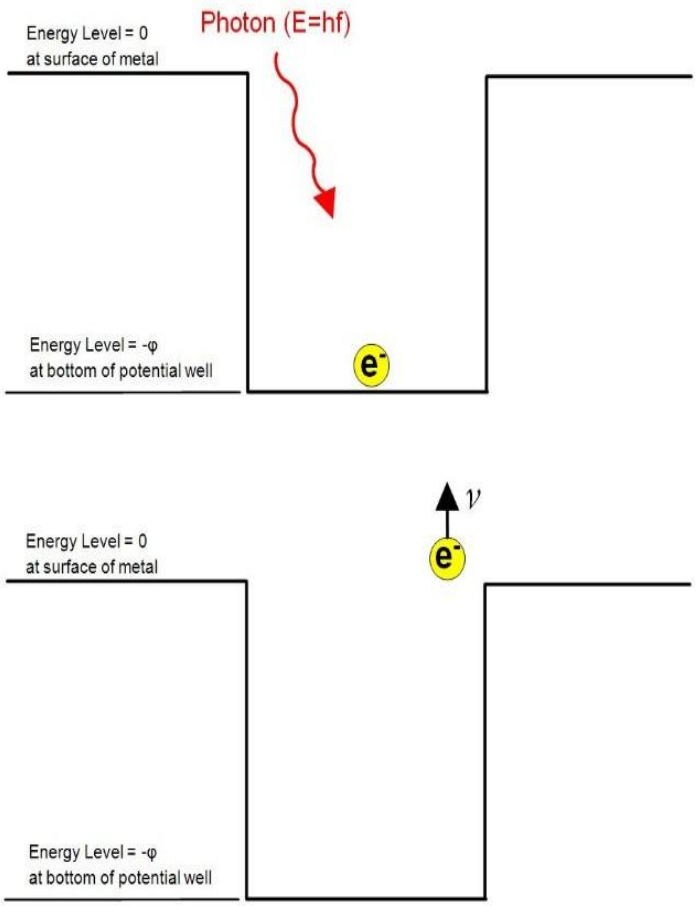


Fig. 1B. Electron absorbing a photon and being emitted from the metal with kinetic energy.

In this experiment, the kinetic energy of the emitted electrons is measured by applying a potential difference (V) across the photodiode, sufficient to just prevent the electrons from reaching the collector plate in the photodiode tube (Figure 2). The voltage is adjusted until the ammeter reads zero. The kinetic energy equals the potential energy, given by the product of the electron charge and the applied potential difference:

$$\mathrm{K} = e\mathrm{V} \tag{3}$$

Substituting equations (2) and (3) into equation (1) gives:

$$\mathrm{eV} = hf - \varphi \tag{4}$$

Solving for the potential difference yields

$$\mathrm{V} = \left(\frac{h}{e}\right) f - \frac{\varphi}{e} \tag{5}$$

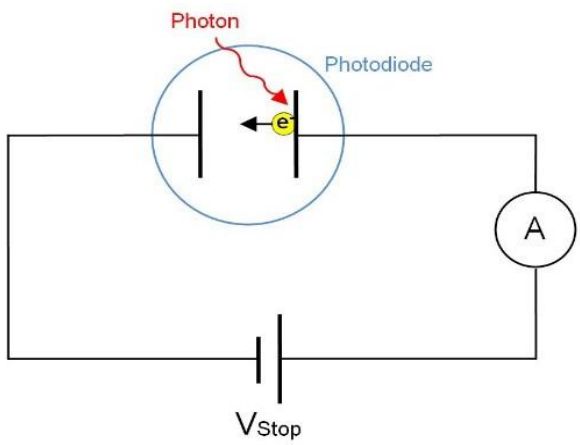


Fig. 2. Photodiode circuit

When light of different frequencies is incident on the metal, electrons are emitted with different kinetic energies, requiring different potential differences to stop them. A plot of V versus f will yield a straight line with a slope of h/e and a y-intercept of φ/e.

By setting the voltage equal zero and solving equation (6) for the frequency, one obtains the x-intercept on the V versus f graph. Only photons with frequencies greater than this threshold can eject electrons from the metal. This threshold is known as the cut off frequency.

$$0 = \left(\frac{h}{e}\right) f_{\mathrm{cutoff}} - \frac{\varphi}{e} \tag{6}$$

$$f_{\mathrm{cutoff}} = \frac{\varphi}{h} \tag{7}$$